



From the Editor's Desk...

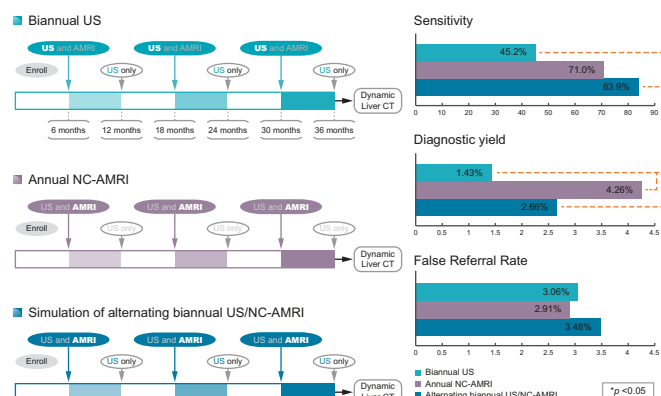
Patrizia Burra*, Frank Tacke, Vlad Ratziu, Stefan Zeuzem, Bruno Sangro, Paolo Angeli

SELECTION OF THE MONTH

Comparison of non-contrast abbreviated MRI vs. US for liver cancer surveillance.

Ultrasonography (US) is the mainstay of surveillance in patients at high risk of developing hepatocellular carcinoma (HCC). However, the sensitivity of US for detecting early-stage HCC is limited and non-contrast-enhanced abbreviated MRI (AMRI) has been proposed as an alternative based on retrospective studies.

Kim, Yoon and coworkers have conducted a prospective study in which 208 patients with an estimated annual risk of HCC greater than 5% went through six rounds of surveillance at 6-month intervals. US was used in every round (980 exams) and AMRI was used in every other round (516 exams) starting at the first one. After 30 months of follow-up, 34 HCCs were detected in 31 patients, and all of them were at the very early (64.5%) or early (35.5%) stages. Compared to biannual US, annual AMRI showed a marginally higher sensitivity (71.0% vs. 45.2%), a significantly higher diagnostic yield (4.26% vs. 1.43%) and a similar false referral rate (2.91% vs. 3.06). A simulation of alternating US and NC-AMRI at 6-month intervals yielded a sensitivity of 83.0%. **This is the first prospective evidence supporting the use of AMRI in surveillance programmes for patients at very high risk of HCC.**



Kim, Yoon *et al.*, 2024.

Comparison of non-contrast abbreviated MRI vs. ultrasound for liver cancer surveillance.

EXPERIMENTAL AND TRANSLATIONAL HEPATOLOGY

Inhibiting inflammatory macrophage activation via EFHD2 decelerates MASLD progression.

Inflammation is considered a key factor in the pathogenesis of metabolic dysfunction-associated steatotic liver disease (MASLD), which determines progression from simple steatosis to metabolic dysfunction-associated steatohepatitis (MASH) and its complications fibrosis and liver cancer. **Fu, Liu, Wu, Chen, Lu, Cao, and coworkers** demonstrate that monocyte-derived macrophages upregulate a molecule termed EF-hand domain family member D2 (EFHD2) in patients with MASH as well as in dietary mouse models of MASH. In mice, specific deletion of EFHD2 in myeloid cells ameliorates typical features of progressive MASH including fibrosis and HCC, due to blunted interferon gamma signalling in macrophages. The authors further

developed a peptide targeting EFHD2 that was effective in reducing MASH progression in mice. These data fuel the expectation that this factor could be pharmacologically targeted to prevent MASLD progression in humans.

Neutralising the obesity-related secreted ER membrane protein complex reduces steatosis.

Endoplasmic reticulum (ER) stress is a feature of metabolically injured hepatocytes in MASLD, and ER stress is modulated via ER membrane protein complex (EMC) and its subunits. **Chen, Wang, Yang, and coworkers** demonstrate that different EMC10 subunits have counteracting activities on MASLD pathogenesis, which has already been linked to obesity. The secreted form of EMC10 (scEMC10) promoted, while its membrane-bound form (mEMC10) suppressed hepatic steatosis in mice. Similar associations were also observed in patients with different stages of

MASLD. Interestingly, neutralisation of scEMC10 via an antibody ameliorated MASLD in mice, suggesting a novel systemic pathway that could be targeted to ameliorate both hepatic as well as adipose tissue manifestations of the metabolic syndrome.

Gut dysbiosis drives liver disease in cystic fibrosis.

A substantial fraction of individuals with cystic fibrosis (CF) will develop CF-related liver disease (CFLD), which is a major contributor to morbidity and mortality in CF. **Bertolini and coworkers** applied sophisticated models of CF in mice, with or without correcting the *CFTR* gene in the intestine, complemented by experiments with antibiotic treatment or faecal microbiota transfer. Restoring the *CFTR* gene in the intestine was sufficient to protect mice from CFLD and antibiotic treatment alone ameliorated hepatic inflammation. These data indicate that **CF-related gut dysbiosis**

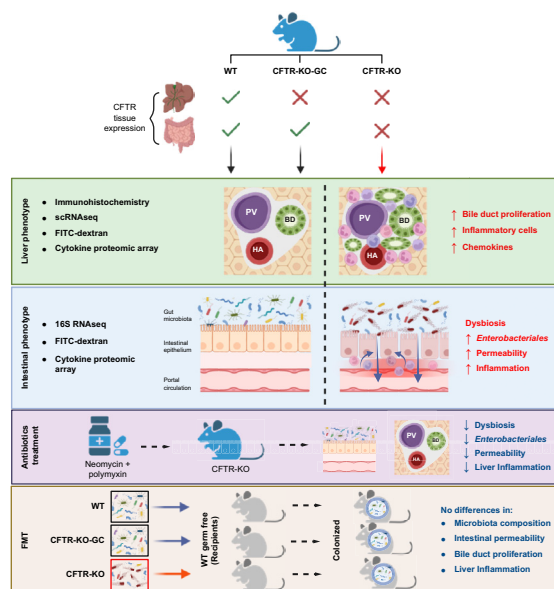


is a major driver of liver disease in CF via increased intestinal permeability and intestinal inflammation. The validation of this concept for pathogenesis of CFLD in humans is eagerly awaited, as this would open new avenues for microbiota modulation as a preventive or therapeutic approach in CFLD.

JNJ-73763989 and bersacapavir treatment in nucleos(t)ide analogue suppressed patients with chronic hepatitis B.

Agarwal and coworkers report data from a double-blind, placebo-controlled, randomised study that enrolled 130 patients with nucleos(t)ide analogue (NA)-sup-

pressed these two predisposing factors. **Chen and coworkers** analysed 21,619 participants from the UK Biobank with data on dietary intake and *PNPLA3*, *TM6SF2* and a 16-variant hepatic steatosis polygenic risk score (PRS) to study their impact on liver fat content (measured by MRI), cT1 (an MRI-based measure of hepatic fibroinflammatory lesions), and liver-related mortality. There was a significant interaction between diet and genetic markers, and this deleterious interaction was stronger in overweight than in lean people. Dietary effects on liver fat content were up to 3.8 times as high in people bearing the *PNPLA3* mutated gene and 3 times higher in the top vs. the bottom quartile of the steatosis PRS. Dietary and genetic predictors were associated with cT1 with the same directionality. Both dietary and genetic predictors were associated with liver-related events and mortality. This study shows that **dietary patterns may influence the severity of hepatic steatosis and of inflammation/fibrosis even more in people at increased genetic risk than in those at low or average risk.** It also suggests greater benefits of dietary modification in individuals with high baseline genetic risk. The authors conclude that patients at increased genetic risk (based on individual variants or PRSs) should be prioritised for intensive lifestyle interventions.



Bertolini et al., 2024.

Gut dysbiosis drives liver disease in cystic fibrosis.

VIRAL HEPATITIS

B cell characteristics define HCV reinfection outcome.

In individuals who are reinfected with HCV, some will develop a persistent infection, while a small proportion repeatedly clear the virus, suggesting natural protection is possible.

Underwood, Gupta, and coworkers examined broad neutralising antibodies and envelope 2-specific memory B cell responses longitudinally in 15 individuals with varied reinfection outcomes. Broad neutralising antibody responses were associated with memory B cell recall, but not with reinfection clearance. Strong evidence of antigen imprinting was found, and the **B cell receptor repertoire showed a high level of clonality with ongoing somatic hypermutation of many clones over subsequent reinfection events.** Single cell transcriptomic analyses showed an activated transcriptomic profile in HCV-specific B cells that rapidly expanded upon reinfection.

pressed HBeAg-negative chronic hepatitis B who received the small-interfering RNA JNJ-3989 (200 mg subcutaneously every 4 weeks) and the capsid assembly modulator bersacapavir (250 mg oral daily) + NA (oral daily; active arm) or placebos (control arm) for 48 weeks. At follow-up Week 24 and 48, no patient achieved the primary endpoint of HBsAg seroclearance. A pronounced on-treatment reduction in mean HBsAg from baseline at Week 48 in the active arm vs. no decline in the control arm was observed. **At follow-up Week 48, reductions from baseline were >1 log₁₀ IU/ml in 81.5% vs. 12.5% of patients in the active and control arms, respectively.**

MASLD

Genetic risk accentuates dietary effects on hepatic steatosis, inflammation and fibrosis.

There is abundant research data linking dietary patterns with MASLD prevalence and genetic polymorphisms with MASLD prevalence and severity. However, we lack understanding of the interactions between

CIRRHOSIS AND LIVER FAILURE

Prospective validation of the EASL management algorithm for AKI in cirrhosis.

The management of acute kidney injury (AKI) in cirrhosis is challenging. The EASL guidelines proposed an algorithm, which had not been validated. In this study, **Ma and coworkers** aimed to prospectively evaluate this algorithm in clinical practice through a prospective cohort study of consecutive hospitalised patients with cirrhosis and AKI. A total of 202 AKI episodes in 139 patients were included. Overall, treatment response was 80%, while renal replacement therapy was required in only 8% of patients. Response to albumin infusion was achieved in one-third of episodes. Among the patients not responding to albumin, most (74%) did not meet the diagnostic criteria of HRS-AKI, with ATN being the most common phenotype. The response rate in patients not meeting HRS-AKI criteria was 70%. Only 30 patients met the diagnostic criteria

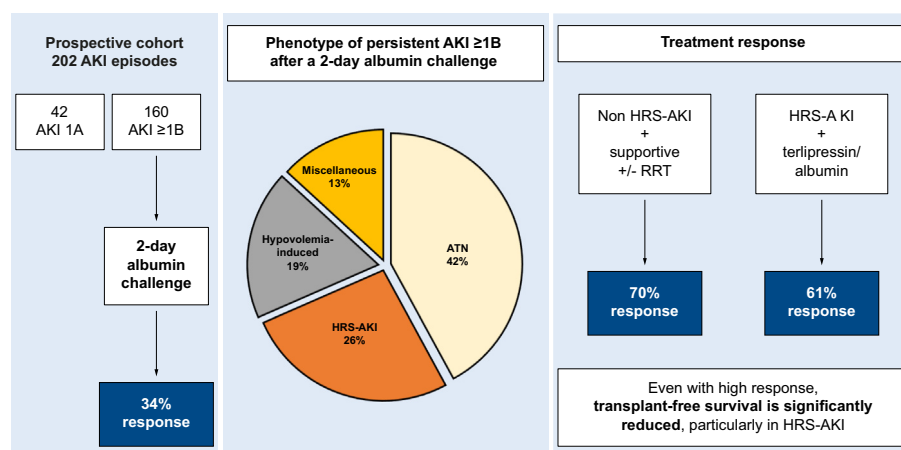
for HRS-AKI, and response rate to terlipressin was 61%. Median time from AKI diagnosis to terlipressin initiation was only 2.5 days. While uNGAL (urinary neutrophil gelatinase-associated lipocalin) could differentiate ATN from other phenotypes, it did not predict response to therapy in HRS-AKI. **Ninety-day transplant-free survival was negatively associated with MELD-Na, ATN and HRS-AKI as well as uNGAL.** Thus, the authors concluded that the application of the EASL AKI algorithm is associated with very good response rates and does not significantly delay initiation of terlipressin therapy.

this retrospective study, **Ollivier-Hourmand, Lebedel and coworkers** aimed to determine factors associated with splanchnic or extra-splanchnic new thrombotic events in this setting. The study included recent non-cirrhotic portal vein thromboses associated with local factors. High and low prothrombotic risk factors, prespecified according to Riport study criteria, were assessed. During follow up, 17 new thrombotic events occurred after a median follow-up of 52 months. **New thrombosis was associated with high risk factors, but inversely related to recanalization and anti-**

associated with local factors, high risk factors for thrombosis are associated with new thrombotic events. Permanent anticoagulation appears beneficial in this high-risk situation.

LIVER TRANSPLANTATION

The National Liver Offering Scheme (NLOS) was introduced in the UK in 2018 to offer livers from deceased donors to patients on the national waiting list based, for most patients, on calculated transplant benefit. **Allen and coworkers** assessed the UK Transplant Registry data on patient registrations and transplants to build survival on the list (M1) and survival post-transplantation (M2) statistical models. A separate cohort of registrations was analysed to simulate what liver allocation would have been under M1, M2 and a transplant benefit score (TBS) model, and to compare these allocations to what had been recorded in the registry. Number of deaths on the waiting list and patient life years were used to compare the different simulation scenarios and to select the optimal allocation model. The TBS was identified as the optimal model to offer donation after brain death livers to adult and large paediatric elective recipients and, in the first 2 years of NLOS, 68% of donation after brain death livers were offered using the TBS to this type of recipient. Monitoring data indicate that mortality on the waiting list post-NLOS significantly decreased compared with pre-NLOS ($p < 0.0001$), and that patient survival post-listing is significantly greater post- vs. pre-NLOS ($p = 0.005$). In the first 2 years of NLOS offering, **waiting list mortality fell while post-transplant survival was not negatively impacted, delivering on the scheme's objectives.**



Ma et al., 2024.

Prospective validation of the EASL management algorithm for AKI in cirrhosis.

Recurrent splanchnic and extra-splanchnic thrombotic events in non-cirrhotic portal vein thrombosis associated with local factors.

One-third of recent non-cirrhotic portal vein thromboses are associated with local factors. The risk of re-thrombosis after anticoagulation withdrawal is unknown. In

coagulation. When a high-risk factor was present, a new thrombotic event occurred in 7.4%, 14.6%, 14.6% and 28.8% of patients at 1, 3, 5 and 7 years under anticoagulants compared to 21.2%, 21.2%, 58% and 58% without anticoagulants. Thus, the authors concluded that in recent non-cirrhotic portal vein thrombosis

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